Lecture 3 Consumer Theory II

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Individual decision-making under certainty

Course outline

We will divide decision-making under certainty into three units:

1. Producer theory
   - Feasible set defined by technology
   - Objective function $p \cdot y$ depends on prices
2. Abstract choice theory
   - Feasible set totally general
   - Objective function may not even exist
3. Consumer theory
   - Feasible set defined by budget constraint and depends on prices
   - Objective function $u(x)$

The consumer problem

Utility Maximization Problem

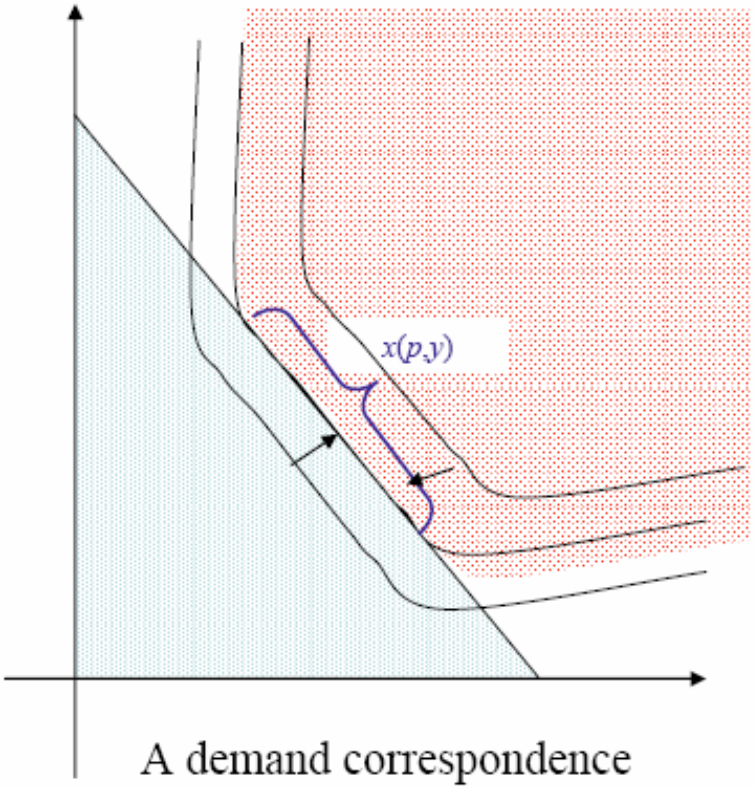
$$\max_{x \in \mathbb{R}^n_+} u(x) \text{ such that } \underbrace{p \cdot x}_{\text{Expenses}} \leq w$$

where p are the prices of goods and w is the consumer's "wealth."

This type of choice set is a budget set

$$B(p, w) \equiv \{x \in \mathbb{R}^n_+ : p \cdot x \leq w\}$$

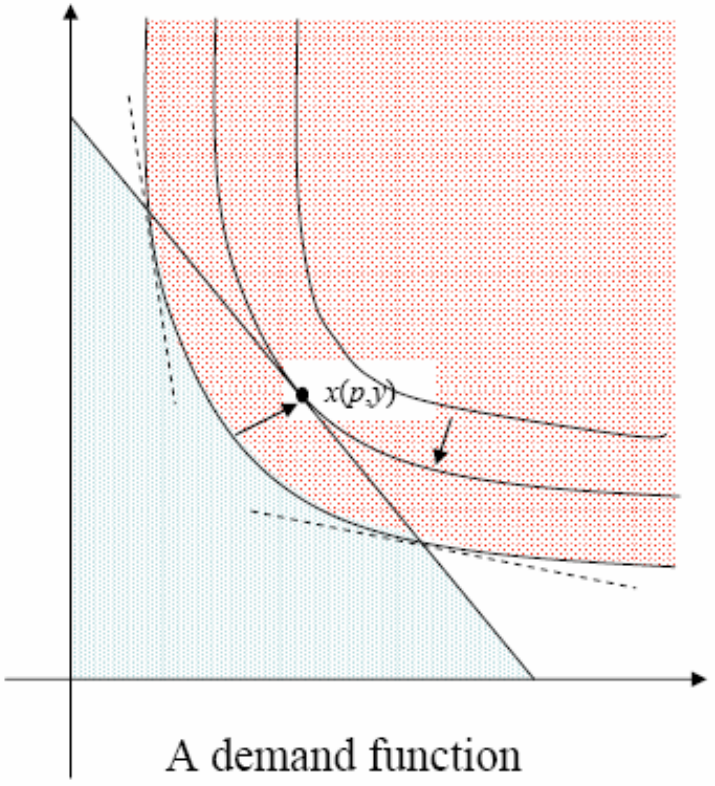
Illustrating the Utility Maximization Problem



A demand correspondence

↓

(Convex $\succsim$)



A demand function
(a single valued set)

↓

(Strictly convex $\succsim$)

Assumptions underlying the UMP

Note that

- Utility function is general (but assumed to exist—a restriction of preferences)
- Choice set defined by linear budget constraint
  - Consumers are price takers
  - Prices are linear
  - Perfect information: prices are all known
- Finite number of goods
  - Goods are described by quantity and price
  - Goods are divisible
  - Goods may be time- or situation-dependent
  - Perfect information: goods are all well understood

Outline

- The utility maximization problem
  - Marshallian demand and indirect utility
  - First-order conditions of the UMP
  - Recovering demand from indirect utility
- The expenditure minimization problem
- Wealth and substitution effects
  - The Slutsky equation
  - Comparative statics properties

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- The utility maximization problem
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Utility maximization problem

The consumer's Marshallian demand is given by correspondence $x\colon \mathbb{R}^n \times \mathbb{R} \rightrightarrows \mathbb{R}^n_+$

$$x(p,w) \equiv \underset{x\in\mathbb{R}^n_+\colon p\cdot x\leq w}{\operatorname{argmax}} u(x) \equiv \underset{x\in B(p,w)}{\operatorname{argmax}} u(x)$$
$$= \left\{x \in \mathbb{R}^n_+ \colon p\cdot x \leq w \text{ and } u(x) = v(p,w)\right\}$$

Resulting indirect utility function is given by

$$v(p,w) \equiv \sup_{x\in\mathbb{R}^n_+\colon p\cdot x\leq w} u(x) \equiv \sup_{x\in B(p,w)} u(x)$$

Properties of Marshallian demand and indirect utility

Theorem

$v(p, w)$ and $x(p, w)$ are homogeneous of degree zero. That is, for all p, w, and $\lambda > 0$,

$$v(\lambda p, \lambda w) = v(p, w) \text{ and } x(\lambda p, \lambda w) = x(p, w).$$

These are "no money illusion" conditions

Proof.

$B(\lambda p, \lambda w) = B(p, w)$, so consumers are solving the same problem. □

Implications of restrictions on preferences: continuity

Theorem

If preferences are continuous, $x(p, w) \neq \varnothing$ for every $p \gg \mathbf{0}$ and $w \geq 0$.

i.e., Consumers choose *something*

Proof.

$B(p, w) \equiv \{x \in \mathbb{R}^n_+ : p \cdot x \leq w\}$ is a closed, bounded set. Continuous preferences can be represented by a continuous utility function $\tilde{u}(\cdot)$, and a continuous function achieves a maximum somewhere on a closed, bounded set. Since $\tilde{u}(\cdot)$ represents the same preferences as $u(\cdot)$, we know $\tilde{u}(\cdot)$ must achieve a maximum precisely where $u(\cdot)$ does. □

Implications of restrictions on preferences: convexity I

Theorem

If preferences are convex, then $x(p, w)$ is a convex set for every $p \gg \mathbf{0}$ and $w \geq 0$.

Proof.

$B(p, w) \equiv \{x \in \mathbb{R}^n_+ : p \cdot x \leq w\}$ is a convex set.
If x, $x' \in x(p, w)$, then $x \sim x'$.
For all $\lambda \in [0, 1]$, we have $\lambda x + (1 - \lambda)x' \in B(p, w)$ by convexity of $B(p, w)$ and $\lambda x + (1 - \lambda)x' \succsim x$ by convexity of preferences. Thus

$$\lambda x + (1 - \lambda)x' \in x(p, w).$$

□

Implications of restrictions on preferences: convexity II

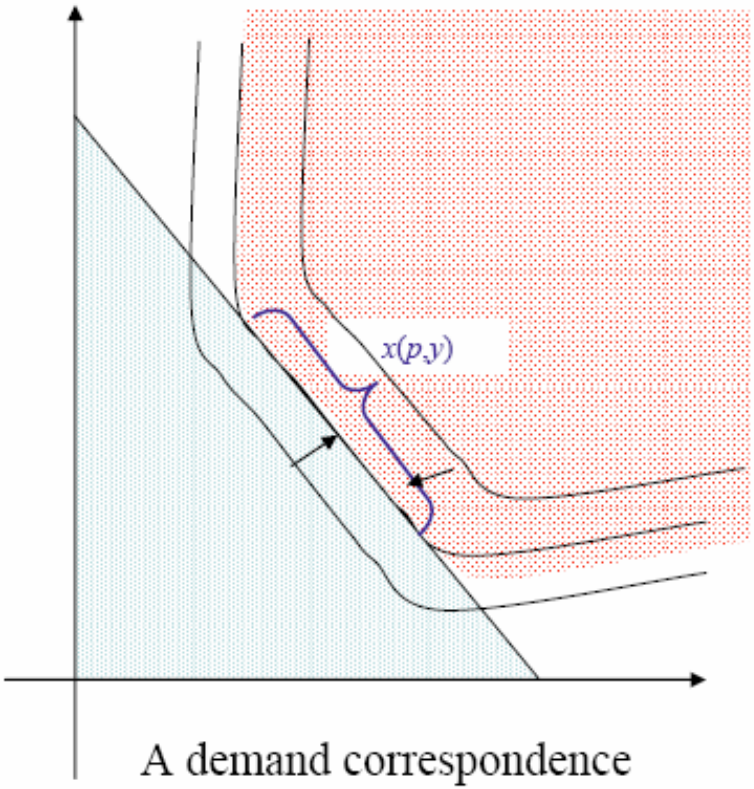
Theorem

If preferences are strictly convex, then $x(p, w)$ is single-valued for every $p \gg \mathbf{0}$ and $w \geq 0$.

Proof.

$B(p, w) \equiv \{x \in \mathbb{R}^n_+ : p \cdot x \leq w\}$ is a convex set.
If x, $x' \in x(p, w)$, then $x \sim x'$. Suppose $x \neq x'$.
For all $\lambda \in (0, 1)$, we have $\lambda x + (1 - \lambda)x' \in B(p, w)$ by convexity of $B(p, w)$ and $\lambda x + (1 - \lambda)x' \succ x$ by convexity of preferences.
But this contradicts the fact that $x \in x(p, w)$. Thus $x = x'$. □

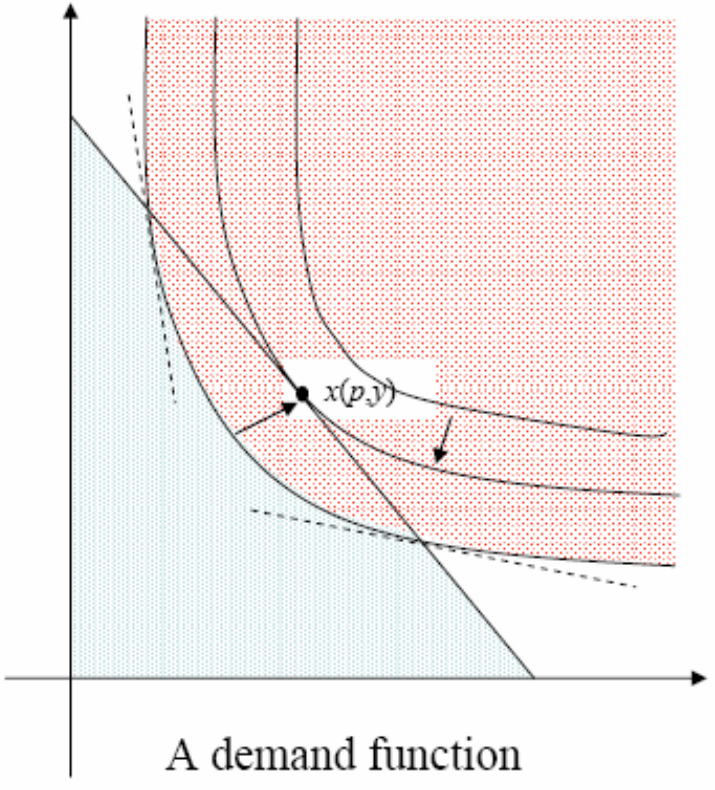
Implications of restrictions on preferences: convexity III



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Implications of restrictions on preferences: non-satiation I

Definition (Walras' Law)

$p \cdot x = w$ for every $p \gg \mathbf{0}$, $w \geq 0$, and $x \in x(p, w)$.

Theorem

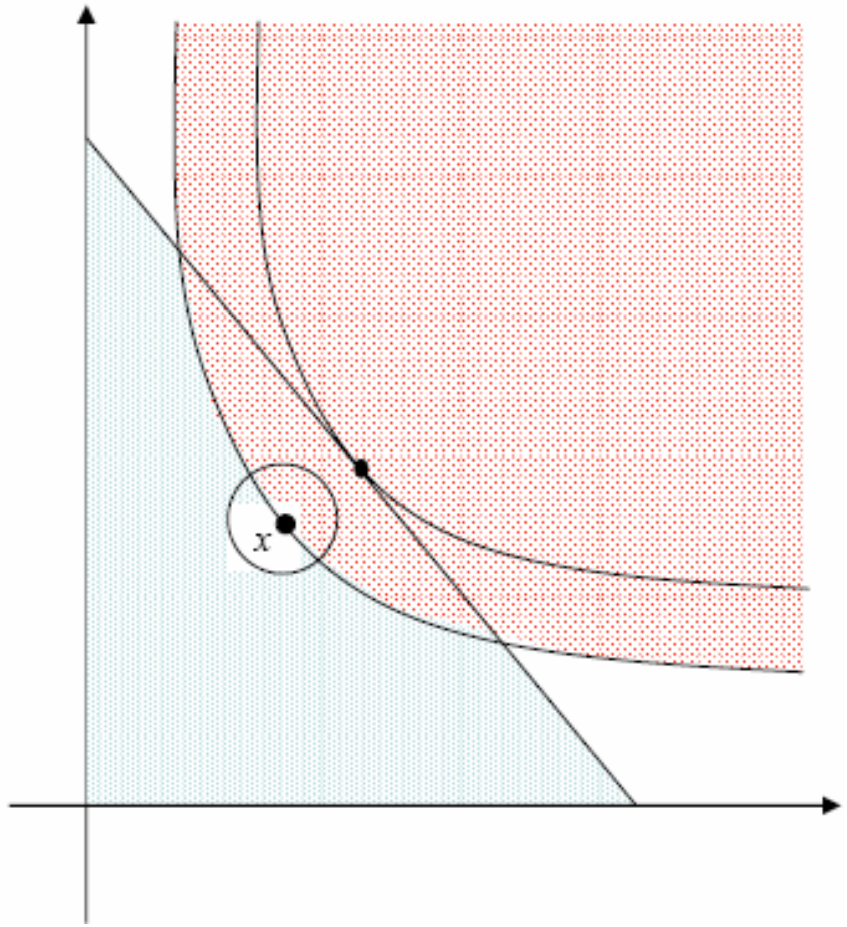
If preferences are locally non-satiated, then Walras' Law holds.

This allows us to replace the inequality constraint in the UMP with an equality constraint

Implications of restrictions on preferences: non-satiation II

Proof.

Suppose that $p \cdot x < w$ for some $x \in x(p, w)$. Then there exists some x' sufficiently close to x with $x' \succ x$ and $p \cdot x' < w$, which contradicts the fact that $x \in x(p, w)$. Thus $p \cdot x = w$. □



Solving for Marshallian Demand

- We use the "Kuhn-Tucker Condtion" to solve the UMP.
- Suppose the problem is

$$\underset{x \in R_n^+}{Max}\, U(x) s.t. p \cdot x \leq w$$

set up the Lagrangian

$$L = U(x) + \lambda(w - p \cdot x)$$

Solving for Marshallian Demand

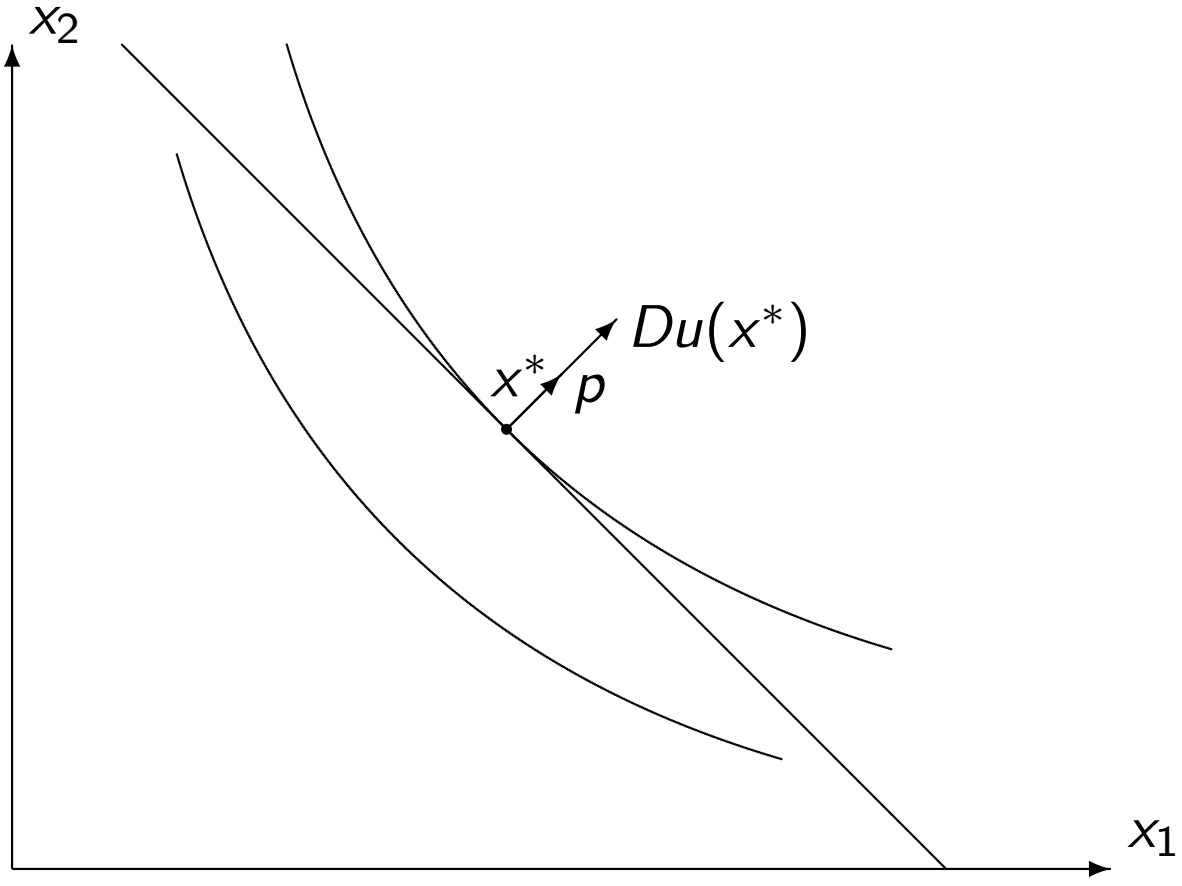
- The first-order condition (F.O.C.) is
- $U_l'(x) \leq \lambda p_l$ and $x \cdot [U_l'(x) - \lambda p_l] = 0$
- So if $x \gg 0$,then we have $U_l'(x) = \lambda p_l$ for all l.

The intuition of the Kuhn-Tucker Condition I

- $\frac{U_l'(x)}{U_k'(x)} = \frac{p_l}{p_k}$ for all l, k such that $x_l > 0$ and $x_k > 0$
- The ratio of marginal utility for any two products are equal to their relative price.
- Consumers trade off utility and cost

Intuition from Kuhn-Tucker conditions II

- The MRS is the (negative) slope of the indifference curve
- Price ratio is the (negative) slope of the budget line



The meaning of the lagrangian multiplier

- Recall that the envelop theorem tells us that the derivative of the value function in some parameter is the derivative of the Lagrangian.
- The value function is the indirect utility function

$$v(p, w) = u(x(p, w))$$

- The lagrangian is

$$L = U(x) + \lambda(w - p \cdot x)$$

- By the E.T., we have $\frac{\partial v}{\partial w} = \lambda$. That is, λ is the "shadow price" of wealth measured in terms of utility.

Recovering Marshallian demand from indirect utility I

To recover the choice correspondence from the value function we typically apply an Envelope Theorem (e.g., Hotelling, Shephard)

- Value function (indirect utility): $v(p, w) \equiv \sup_{x \in B(p,w)} u(x)$
- Lagrangian: $\mathcal{L} \equiv u(x) + \lambda(w - p \cdot x) + \mu \cdot x$

By the ET

$$\frac{\partial v}{\partial w} = \frac{\partial \mathcal{L}}{\partial w} = \lambda$$
$$\frac{\partial v}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial p_i} = -\lambda x_i$$

We can combine these, dividing the second by the first...

Recovering Marshallian demand from indirect utility II

Roy's identity

$$x_i(p, w) = -\frac{\frac{\partial v(p,w)}{\partial p_i}}{\frac{\partial v(p,w)}{\partial w}}.$$

We can think of this a little bit like "$\frac{\partial v}{\partial w} = -\frac{\partial v}{x_i \partial p_i}$"

Here we showed Roy's identity as an application of the ET; the notes give an entirely different proof that relies on the expenditure minimization problem

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Why we need another "problem"

We would like to characterize "important" properties of Marshallian demand $x(\cdot, \cdot)$ and indirect utility $v(\cdot, \cdot)$

- Unfortunately, this is harder than doing so for $y(\cdot)$ and $\pi(\cdot)$
- Difficulty arises from the fact that in UMP parameters enter feasible set rather than objective

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Consider an price increase for one good (apples)

1. Substitution effect: Apples are now relatively more expensive than bananas, so I buy fewer apples
2. Wealth effect: I feel poorer, so I buy _____ (more? fewer?) apples

Wealth effect and substitution effects could go in opposite directions $\implies$ can't easily sign the change in consumption

Isolating the substitution effect

We can isolate the substitution effect by "compensating" the consumer so that her maximized utility does not change

If maximized utility doesn't change, the consumer can't feel richer or poorer; demand changes can therefore be attributed entirely to the substitution effect

Isolating the substitution effect

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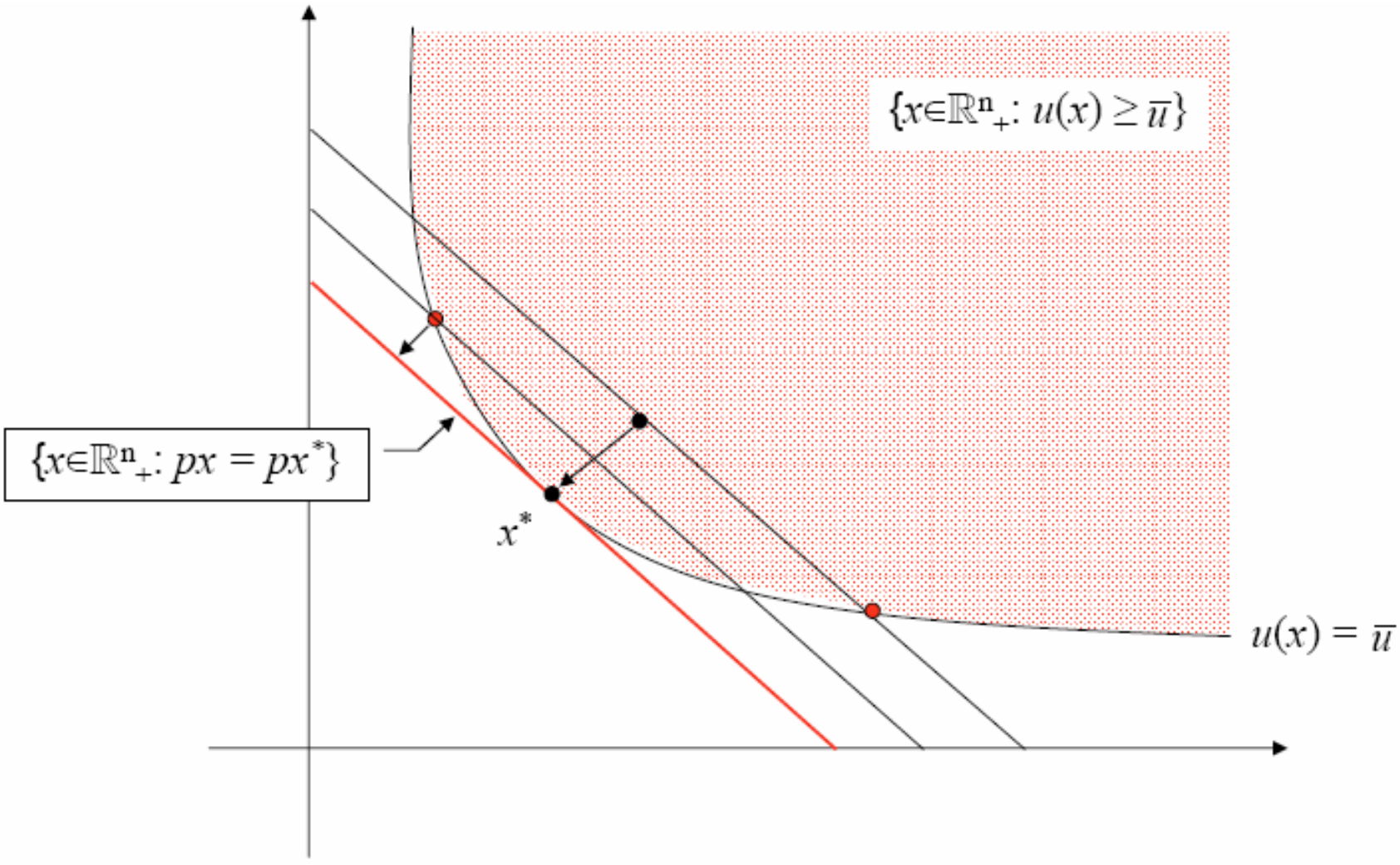
If maximized utility doesn't change, the consumer can't feel richer or poorer; demand changes can therefore be attributed entirely to the substitution effect

Expenditure Minimization Problem

$$\min_{x \in \mathbb{R}^n_+} p \cdot x \text{ such that } u(x) \geq \bar{u}.$$

i.e., find the cheapest bundle at prices p that yield utility at least $\bar{u}$

Illustrating the Expenditure Minimization Problem



Expenditure minimization problem

The consumer's Hicksian demand is given by correspondence $h\colon \mathbb{R}^n \times \mathbb{R} \rightrightarrows \mathbb{R}^n$

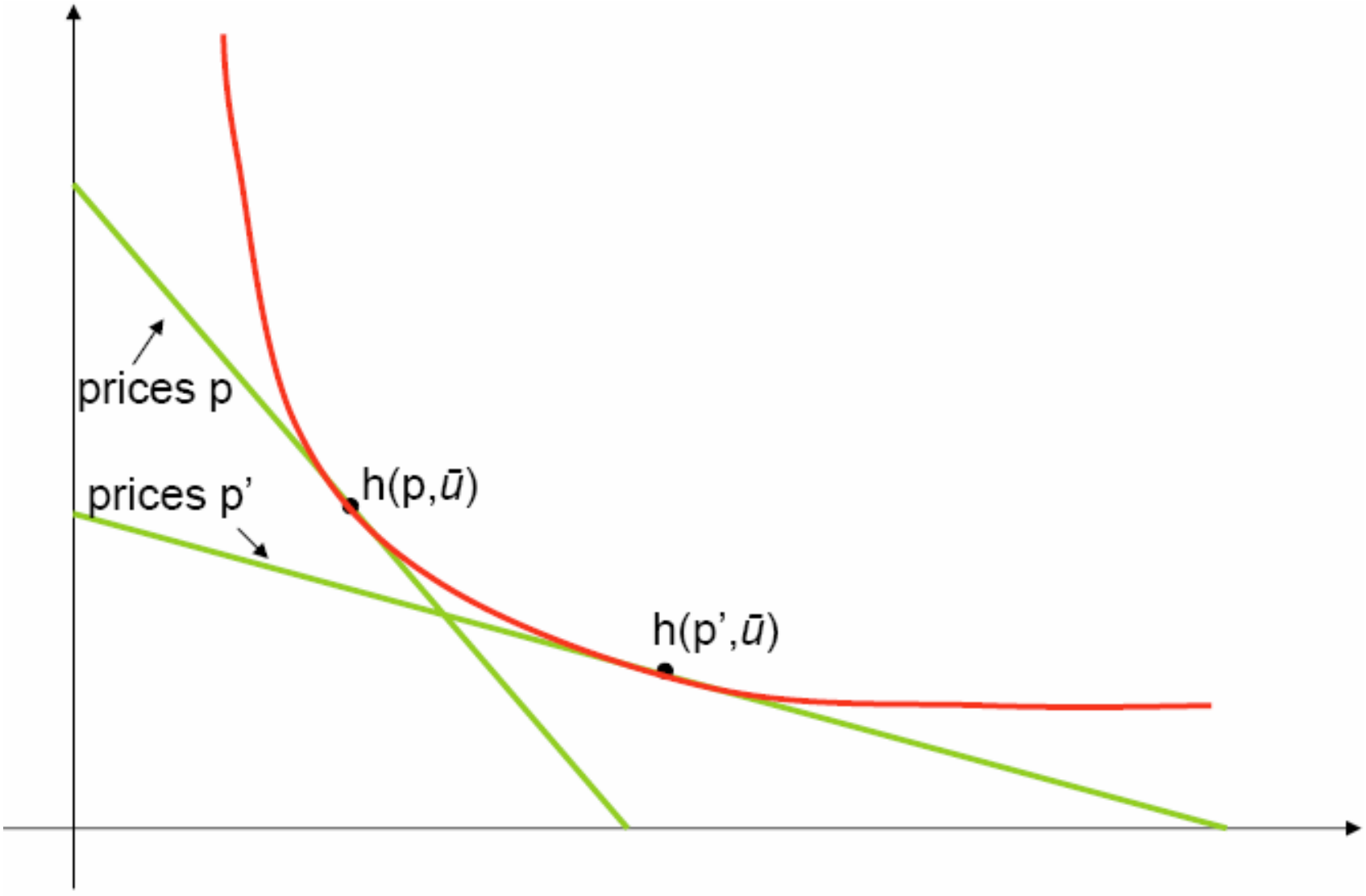
$$h(p,\bar{u}) \equiv \underset{x\in\mathbb{R}^n_+ :\, u(x)\geq\bar{u}}{\operatorname{argmin}} \; p\cdot x$$
$$= \{x \in \mathbb{R}^n_+ : u(x) \geq \bar{u} \text{ and } p\cdot x = e(p,\bar{u})\}$$

Resulting expenditure function is given by

$$e(p,\bar{u}) \equiv \min_{x\in\mathbb{R}^n_+ :\, u(x)\geq\bar{u}} p\cdot x$$

Note we have used min instead of inf assuming conditions (listed in the notes) under which a minimum *is* achieved

Illustrating Hicksian demand



Relating Hicksian and Marshallian demand I

Theorem ("Same problem" identities)

Suppose $u(\cdot)$ is a utility function representing a continuous and locally non-satiated preference relation $\succsim$ on $\mathbb{R}^n_+$. Then for any $p \gg \mathbf{0}$ and $w \geq 0$,

1. $h\big(p, v(p,w)\big) = x(p,w)$,
2. $e\big(p, v(p,w)\big) = w$;

and for any $\bar{u} \geq u(\mathbf{0})$,

3. $x\big(p, e(p,\bar{u})\big) = h(p,\bar{u})$, *and*
4. $v\big(p, e(p,\bar{u})\big) = \bar{u}$.

For proofs see notes (cumbersome but relatively straightforward)

Relating Hicksian and Marshallian demand II

These say that UMP and EMP are fundamentally solving the same problem, so:

- If the utility you can get with wealth w is $v(p, w)$...
  - To achieve utility $v(p, w)$ will cost at least w
  - You will buy the same bundle whether you have w to spend, or you are trying to achieve utility $v(p, w)$
- If it costs $e(p, \bar{u})$ to achieve utility $\bar{u}$...
  - Given wealth $e(p, \bar{u})$ you will achieve utility at most $\bar{u}$
  - You will buy the same bundle whether you have $e(p, \bar{u})$ to spend, or you are trying to achieve utility $\bar{u}$

Properties of Hicksian demand and expenditure I

As in our discussion of the single-output CMP:

- $e(p, \bar{u}) = p \cdot h(p, \bar{u})$ (adding up)
- $e(\cdot, \bar{u})$ is homogeneous of degree one in p
- $h(\cdot, \bar{u})$ is homogeneous of degree zero in p
- If $e(\cdot, \bar{u})$ is differentiable in p, then $\nabla_p e(p, \bar{u}) = h(p, \bar{u})$ (Shephard's Lemma)
- $e(\cdot, \bar{u})$ is concave in p
- If $h(\cdot, \bar{u})$ is differentiable in p, then the matrix $D_p h(p, \bar{u}) = D_p^2 e(p, \bar{u})$ is symmetric and negative semidefinite
- $e(p, \cdot)$ is nondecreasing in $\bar{u}$
- Rationalizability condition...

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Relating (changes in) Hicksian and Marshallian demand

Assuming differentiability and hence single-valuedness, we can differentiate the ith row of the identity

$$h(p, \bar{u}) = x\big(p, e(p, \bar{u})\big)$$

in p_j to get

$$\frac{\partial h_i}{\partial p_j} = \frac{\partial x_i}{\partial p_j} + \frac{\partial x_i}{\partial w} \underbrace{\frac{\partial e}{\partial p_j}}_{=h_j=x_j}$$

$$\frac{\partial h_i}{\partial p_j} = \frac{\partial x_i}{\partial p_j} + \frac{\partial x_i}{\partial w} x_j$$

The Slutsky equation I

Slutsky equation

$$\underbrace{\frac{\partial x_i(p,w)}{\partial p_j}}_{\text{total effect}} = \underbrace{\frac{\partial h_i\big(p,u(x(p,w))\big)}{\partial p_j}}_{\text{substitution effect}} - \underbrace{\frac{\partial x_i(p,w)}{\partial w}x_j(p,w)}_{\text{wealth effect}}$$

for all i and j.

In matrix form, we can instead write

$$\nabla_p x = \nabla_p h - (\nabla_w x)x^\top.$$

The Slutsky equation II

Setting $i = j$, we can decompose the effect of an an increase in p_i

$$\frac{\partial x_i(p,w)}{\partial p_i} = \frac{\partial h_i\big(p, u(x(p,w))\big)}{\partial p_i} - \frac{\partial x_i(p,w)}{\partial w} x_i(p,w)$$

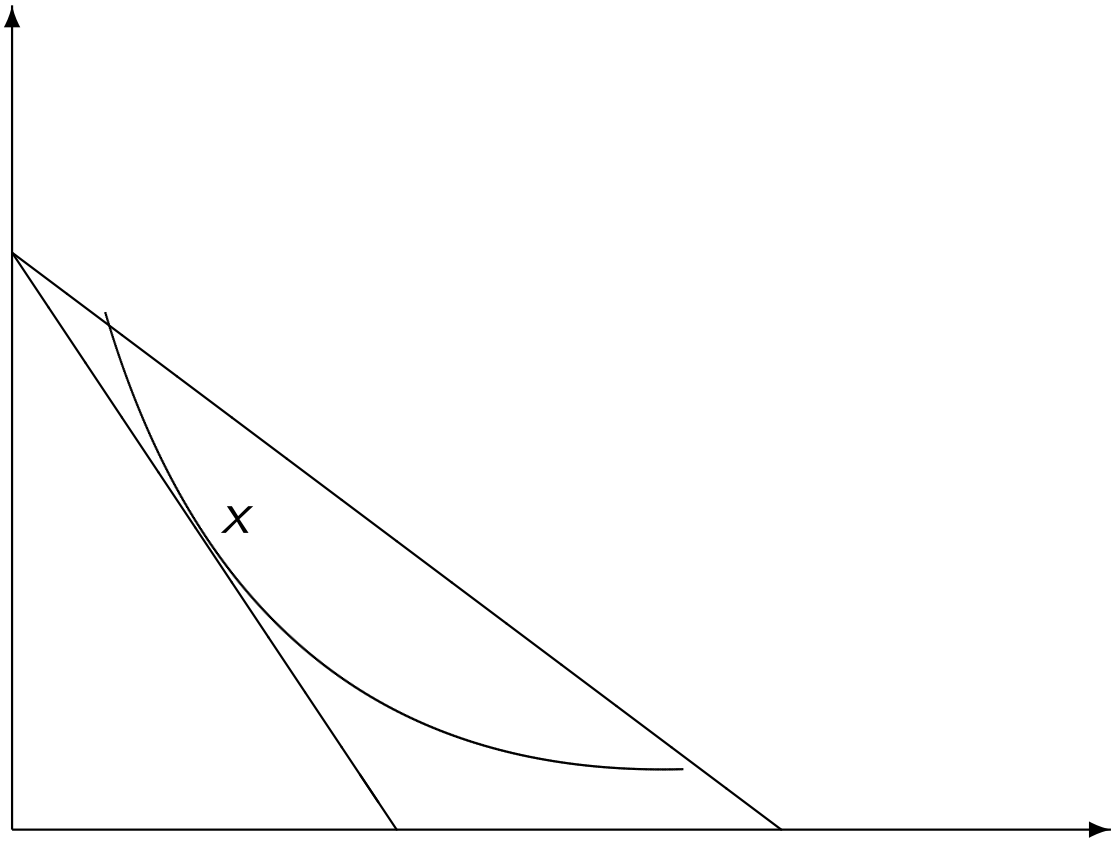
An "own-price" increase...

1. Encourages consumer to substitute away from good i
   - $\frac{\partial h_i}{\partial p_i} \leq 0$ by negative semidefiniteness of Slutsky matrix
2. Makes consumer poorer, which affects consumption of good i in some indeterminate way
   - Sign of $\frac{\partial x_i}{\partial w}$ depends on preferences

Illustrating wealth and substitution effects

Following a decrease in the price of the first good...

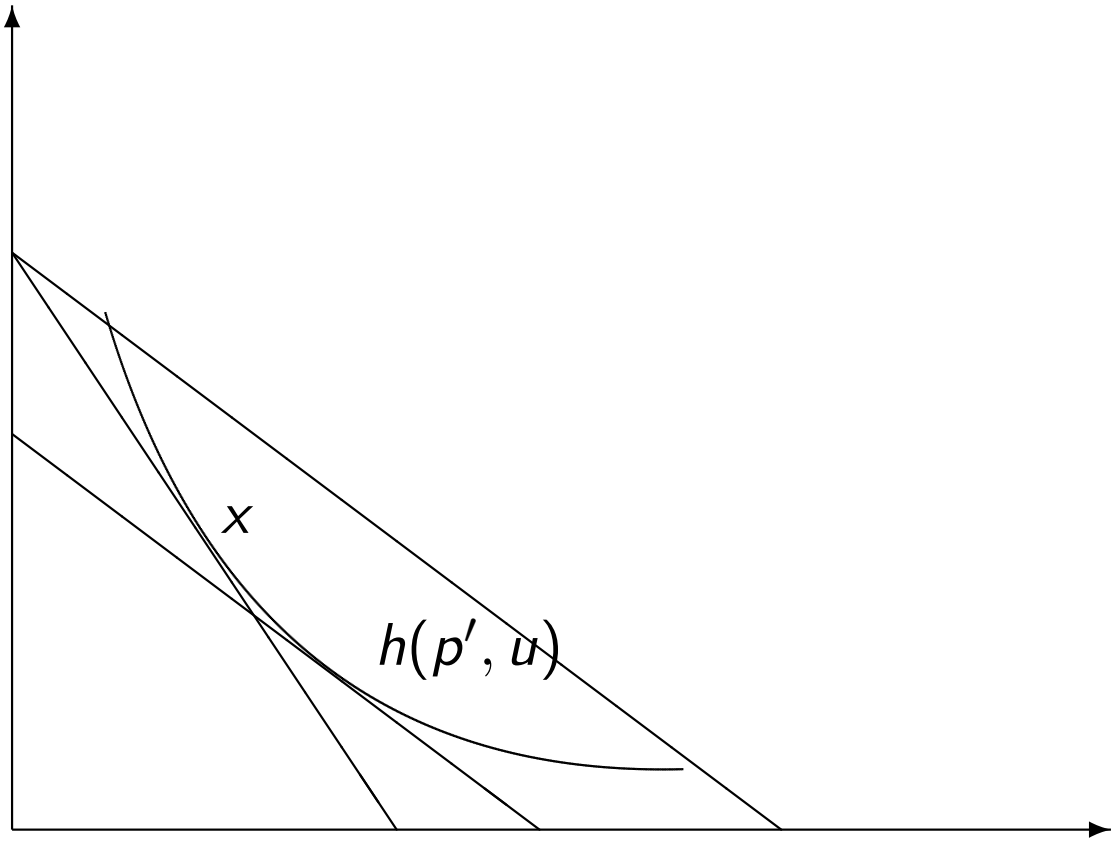
- Substitution effect moves from x to h
- Wealth effect moves from h to x'



Illustrating wealth and substitution effects

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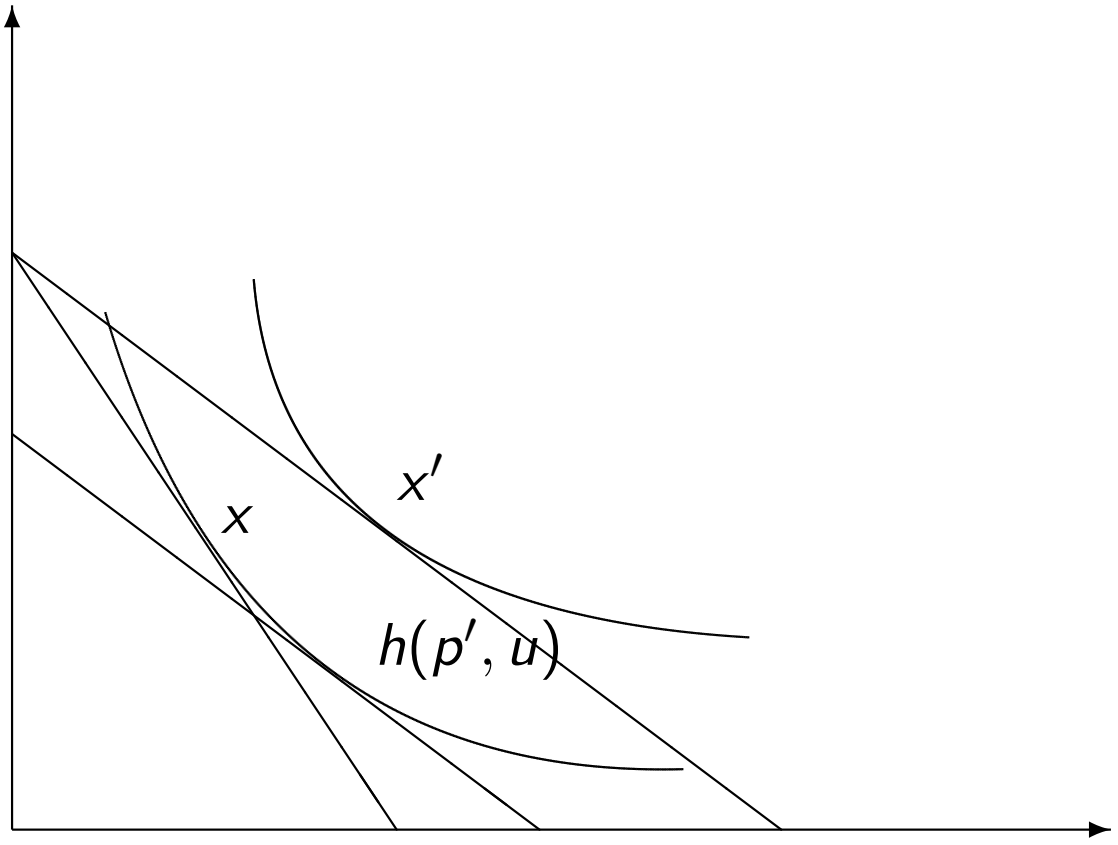
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Illustrating wealth and substitution effects

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Marshallian response to changes in wealth

Definition (Normal good)

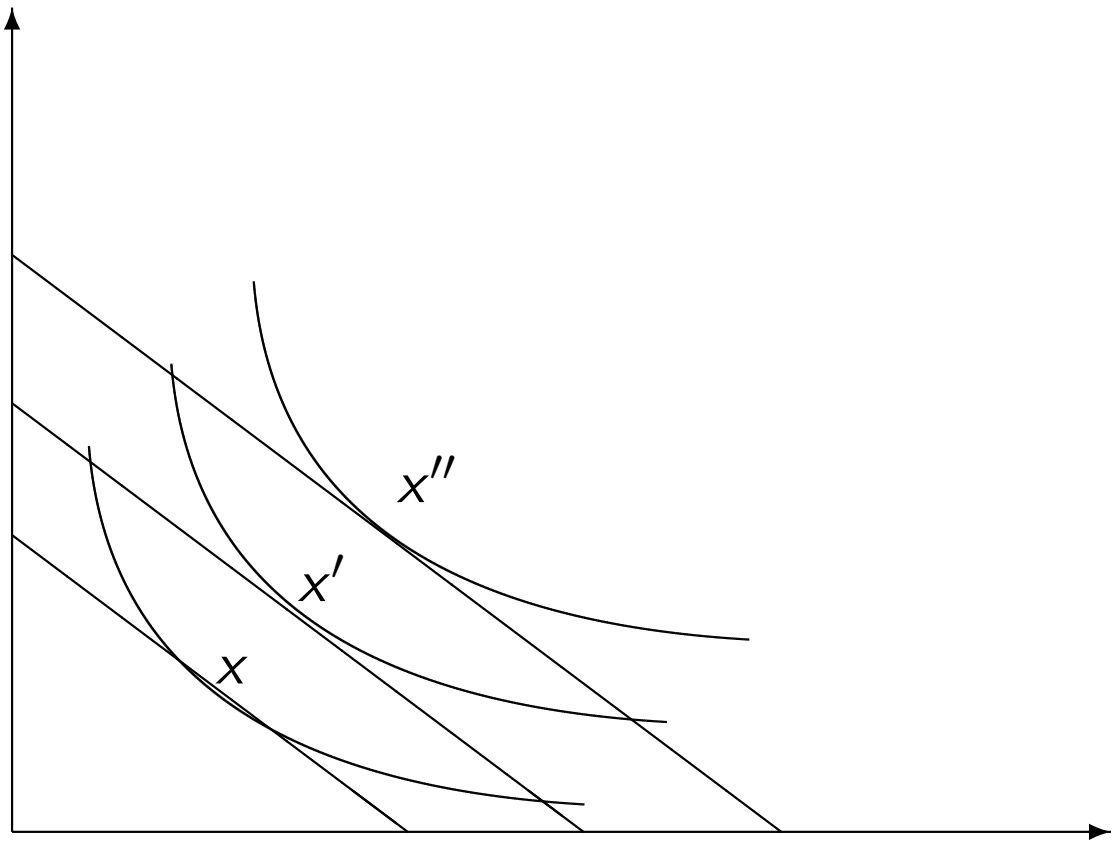
Good i is a normal good if $x_i(p, w)$ is increasing in w.

Definition (Inferior good)

Good i is an inferior good if $x_i(p, w)$ is decreasing in w.

Graphing Marshallian response to changes in wealth

- Engle curves show how Marshallian demand moves with wealth (locus of $\{x, x', x'', \dots\}$ below)
- In this example, both goods are normal (x_i increases in w)



Marshallian response to changes in own price

Definition (Regular good)

Good i is a regular good if $x_i(p, w)$ is decreasing in p_i.

Definition (Giffen good)

Good i is a Giffen good if $x_i(p, w)$ is increasing in p_i.

Potatoes during the Irish potato famine are the canonical example (and probably weren't *actually* Giffen goods)

Marshallian response to changes in own price

Definition (Regular good)

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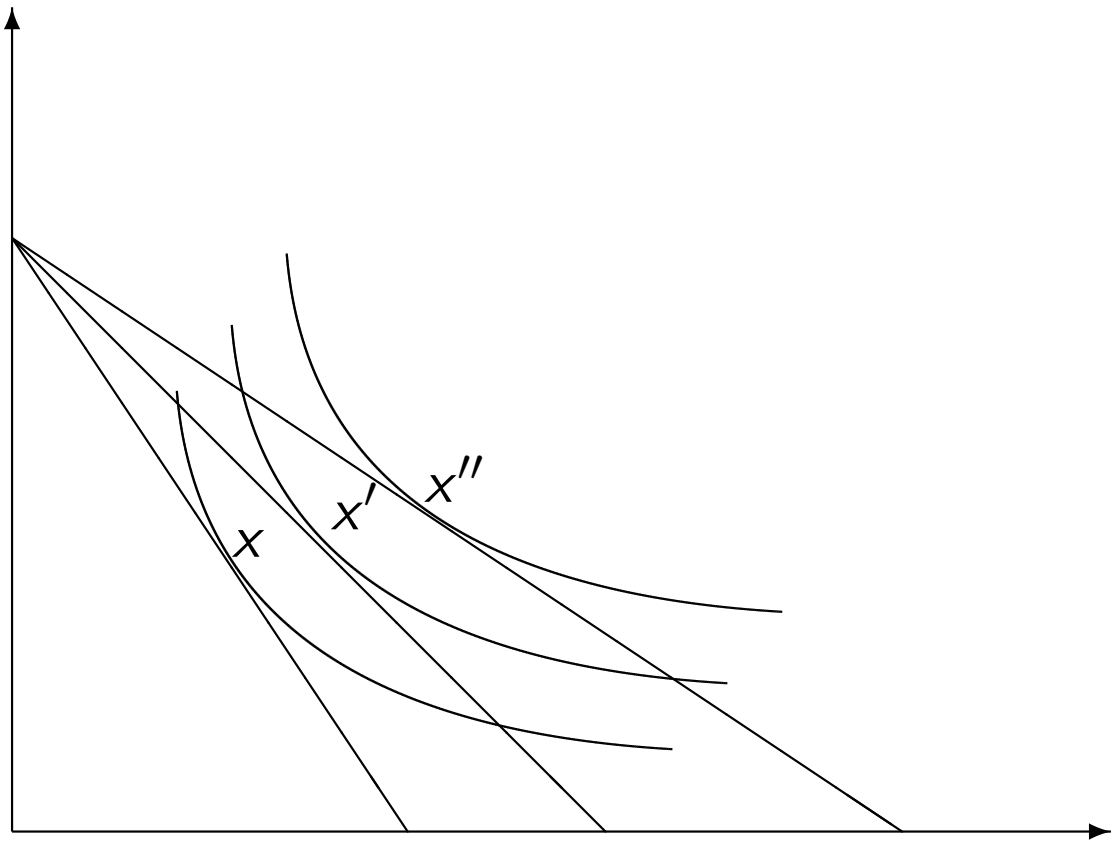
Potatoes during the Irish potato famine are the canonical example (and probably weren't *actually* Giffen goods)

By the Slutsky equation (which gives $\frac{\partial x_i}{\partial p_i} = \frac{\partial h_i}{\partial p_i} - \frac{\partial x_i}{\partial w} x_i$ for $i = j$)

- Normal $\implies$ regular
- Giffen $\implies$ inferior

Graphing Marshallian response to changes in own price

- Offer curves show how Marshallian demand moves with price
- In this example, good 1 is regular and good 2 is a gross complement for good 1



Marshallian response to changes in other goods' price

Definition (Gross substitute)

Good i is a gross substitute for good j if $x_i(p, w)$ is increasing in p_j.

Definition (Gross complement)

Good i is a gross complement for good j if $x_i(p, w)$ is decreasing in p_j.

Gross substitutability/complementarity is not necessarily symmetric

Hicksian response to changes in other goods' price

Definition (Substitute)

Good i is a substitute for good j if $h_i(p, \bar{u})$ is increasing in p_j.

Definition (Complement)

Good i is a complement for good j if $h_i(p, \bar{u})$ is decreasing in p_j.

Substitutability/complementarity is symmetric

In a two-good world, the goods must be substitutes (*why?*)